Executive Summary: Charger Utilisation and Efficiency Analysis

Capacity far exceeds demand.

Chargers are unused 5.4% of the time and sit idle for 22.7 hours out of every 24. Even the busiest hour of the week (Monday 11:00) has chargers in use only 11.5% of the time.

Demand is concentrated at midday on weekdays.

Utilisation peaks about 10:00 and 14:59 (11.2% on weekdays, 8.4% at weekends) and falls to 1.2% overnight. That window delivers 40.7% of daily energy, and network load peaks at 1,882kW on weekdays, 7.4% of the 25,600kW installed.

Congestion is almost non-existent.

Only 14 of 33 stations ever have every charger busy at the same moment, and the highest share of time for any station is 0.061%, about 5 hours a year. The nine 4-charger stations come closest, led by STN_030 and STN_022 in London.

Station size and charger type explain the differences between chargers.

Every station received about 13.9 sessions a day regardless of size, so utilisation falls from 8.5% at 4-charger stations to 4.2% at 8-charger stations, and 50kW chargers average 7.4% against 3.2% for 300kW chargers. Together, these two factors explain 99.7% of the differences between chargers, and differences within a group do not repeat over time (correlation -0.09), so rankings are meaningful for groups and not for individual chargers.

Utilisation is not earning.

The busiest group (50kW chargers at 4-charger stations) earns £67 per charger per day, while 300kW chargers at the same stations earn £153.

Two data issues were found and handled.

8.3% of sessions overlap an earlier session on the same charger (in 99.8% of cases, another charger at that station was free, pointing to an allocation or logging problem and not congestion), and charger installation dates are unreliable. The dataset also appears to be simulated, so the results may not carry over to a real network.

Capacity Optimisation Recommendations

1. **Do not add chargers now.** Covering demand 99.9% of the time would need about 101 of the 199 installed chargers (70 for 99% of the time, 127 for 99.9%).
2. **Watch the nine 4-charger stations first.** They are the only ones near congestion and the first candidate for explosion if demand per station grows.
3. **Right-size the 8-charger stations and plan 4-5 chargers for new sites.** Most 8-charger stations need about 3 chargers at the 99.9% standard. The surplus (about 98 chargers, range 72-129) is indicative only, because it assumes demand stays at about 13.9 sessions per station per day and leaves no buffer for maintenance or faults.
4. **Review the need for 300kW units before buying more.** They are 17.6% of chargers and 41% of installed power but deliver 33% of the energy. They earn more per charger, so the decision needs capital-cost data this dataset doesn't include.
5. **Pilot a change to the peak-price window.** Only 23.6% of energy falls in the 16:00-20:59 premium window, against 40.7% at the midday peak. No price response was detected earlier, so test before relying on it.
6. **Fix charger allocation.** So each goes to a free charger, which would remove the 8.3% of overlapping sessions.
7. **Rank groups, not individual chargers.** Since differences between chargers of the same type and station size are noise.